1. INTRODUCTION

This User Manual provides general installation and use information regarding the Marelli Motori regulator whose code is reported on the cover and inside the document, mounted on Marelli Motori generator types indicated in Chap. 3.
Before starting the generator and performing any type of regulation operation, carefully and completely read all of the instructions contained in this manual.
IMPORTANT NOTE: This User Manual not intended to cover all of the possible application or installation variations nor to provide data or information to support every possible contingency. The connection diagrams provided with the generator, its Use and Maintenance Manual and any additional information provided by qualified Marelli Motori technical personnel supplements and completes this manual.
In particular, the diagrams reported in this document provide only an example of the connection and operation mode for the device; these do not cover all possible application cases and are not a substitute for the connection diagrams normally provided with the generator.
If further application information becomes necessary, please contact Marelli Motori Services.

2. SAFETY PRECAUTIONS



ATTENTION: DO NOT TOUCH THE CONTROL MODULE WHEN IT IS UNDER POWER.
When the control module is under power (or the machine is in rotation) the upper part of the device (connection side) and all parts electrically connected to it contain lethal voltage. There could be also components with high temperatures that are dangerous in the event of direct contact during normal operation on the component module.



Any operation on the cabling and/or mechanical installation of the regulator must be performed by qualified and informed personnel, with the generator halted and making sure that a sufficient amount of time has passed for the regulator components to have cooled down to a temperature that is not dangerous to personal safety.



Each calibration operation must be performed with the machine running without a load by qualified and informed personnel using the proper tools to guarantee electrical safety (for example, insulated screwdrivers) and wearing all of the necessary personal safety devices (especially safety glasses and protective gloves).

Marelli Motori denies any responsibility for damage to the regulator, system or people, for loss of profit or money, or for system shutdowns caused by the failure to observe the safety and/or installation/use instructions reported in this User Manual.

3. APPLICATION

The M31FA600A MEC-20 voltage regulator is suitable for synchronous generators constructed by Marelli Motori.
NOTE: For information on compatibility with different sizes or series of synchronous generators other than those indicated above, contact Marelli Motori Services.

4. TECHNICAL SPECIFICATIONS

The M31FA600A voltage regulator is a micro-processor device with pulse-width modulation (PWM) output stage, and can be calibrated using potentiometers. The circuit board is resin-bonded in order to maintain a high degree of operational reliability even in difficult environmental conditions (high levels of humidity, dust, a salty atmosphere) and in the presence of vibrations.

4.1. Power supply

Type	1-phase
From	❖ Mains ❖ Auxiliary winding ❖ Permanent magnet generator (PMG)
Range	From 170 to 277Vac ± 10% (@ 50 – 400Hz)
Self-excitation	5Vac

4.2. Sensing

Voltage	❖ 1-phase, 2 terminals ❖ 3-phases, 3 terminals ❖ From 220Vac to 480Vac ± 10%, @ 50/60Hz
Current	From 0 to 1Aac (@ 50/60Hz)

4.3. Analogue input

Range 1	From -5 to +5Vdc for a variation from -15% to +15% of the voltage reference
Range 2	Suitable for connection to Power Factor Regulator M50FA400A

4.4. Excitation field data

Resistance	From 3Ω to 20Ω
Continuity	From 0 to 7 Adc
Forcing 10s	From 0 to 15 Adc

4.5. Functions

Soft start	Ramp time adjustable from 0 to 60s
Parallel between generators	❖ Type: Reactive droop compensation ❖ Droop: from 0 to 15%
Overexcitation limiting device	❖ Type: Inverse time characteristic ❖ 1 maximum fixed threshold of 15A, above which de-energizing occurs (after about 10s) ❖ 1 minimum settable threshold, above which the limiting device operate according to the characteristic ❖ 1 open collector output per signal
Under frequency limiting device	❖ Corner frequency adjustable from 40 to 60 Hz ❖ Zero Volt frequency adjustable from 0 to 40 Hz

4.6. Accuracy

From 0 to 100% load	±0.5% @ PF 0.8 - balanced and non deforming load and constant frequency
With ±4% engine governing	±1% @ steady state conditions for load and speed
Steady state	±0.1% @ at constant load and frequency
Thermal drift	±0.5% for a change of 30°C at T_{AMB} in 10 minutes
V/Hz	±2%

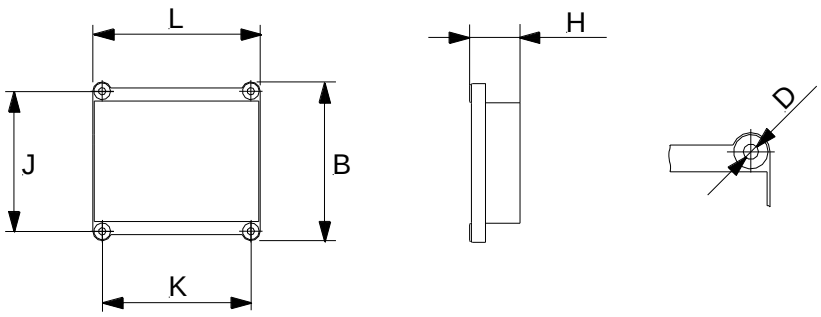
4.7. Environment

Operating temperature	From -30°C to +70°C
Storage	From -40°C to +80°C

4.8. Dimensions

L	180mm
B	170mm
K	160mm
J	150mm
H	40mm
D	6.5mm

Weight	760g



5. TYPE TESTS

5.1. EMC - Emissions

Emission:	**Reference standard EN 61000-6-3**	
Test specifications	*Environmental Phenomena*	*Result*
EN 55022	Conducted disturbance	Complies
EN 55022	Radiated disturbance	Complies
EN 61000-3-2	Harmonic current emissions	Complies
EN 61000-3-3	Voltage fluctuations and flicker	Complies

5.2. EMC - Immunity

Immunity:	**Reference standard EN 61000-6-2**	
Test specifications	*Environmental Phenomena*	*Result*
EN 61000-4-2	Electrostatic discharge	Complies
EN 61000-4-3	Radiated electromagnetic field	Complies
EN 61000-4-4	Electrical fast transients	Complies
EN 61000-4-5	Surge	Complies
EN 61000-4-6	Injected currents	Complies
EN 61000-4-8	Power frequency magnetic field	N.A. (+)
EN 61000-4-11	Dips/short interruptions	Complies

(+) *Apparatus does not contain devices susceptible to magnetic fields*

5.3. Climatic

Reference standard DNV No. 2.4 – 2006		
Test specifications	*Environmental Phenomena*	*Result*
Class: C (-25°C / +55°C) Standard IEC 60068-2-2	Dry Heat	Complies
Class: C (-25°C / +55°C) Standard IEC 60068-2-2	Cold	Complies
Class: C (-25°C / +55°C / 100% R.H.) Standard IEC 60068-2-30	Damp Heat	Complies

5.4. Vibrations

Reference standard DNV No. 2.4 – 2006		
Test specifications	*Environmental Phenomena*	*Result*
Class: B Standard IEC 60068-2-6	Vibration	Complies

6. ACCESSORIES

Description	Code
Fuse Ultra rapid, ceramic, 10 A - 500 V	963823010
External potentiometer 10 kΩ - 2 W	963824425

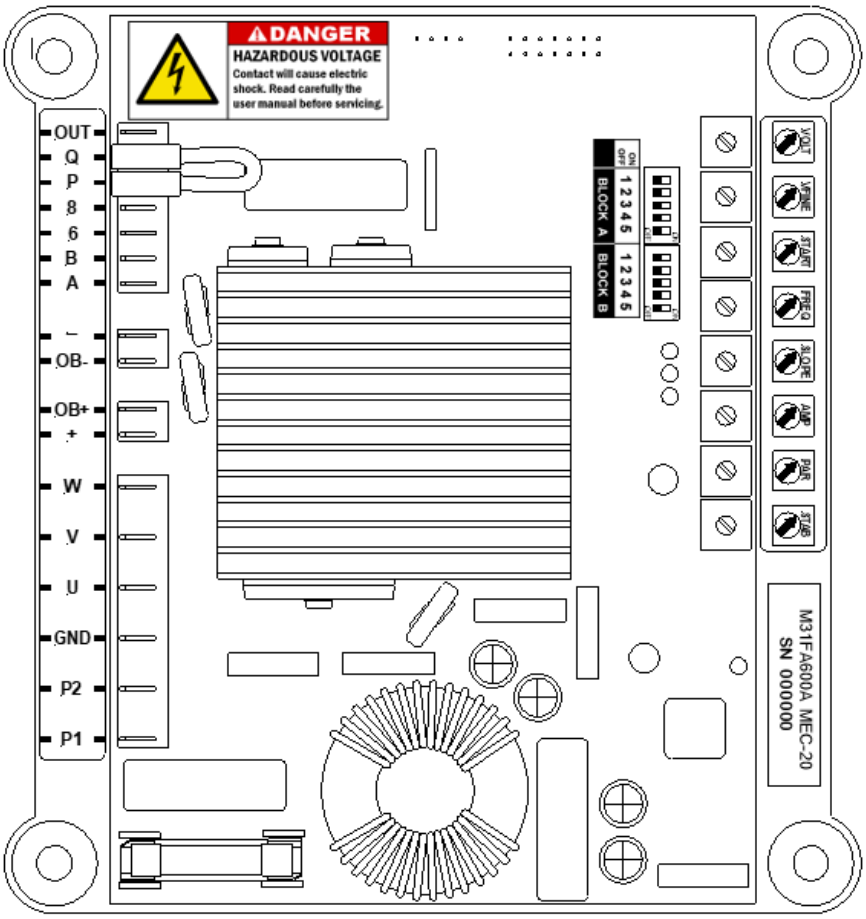
7. COMPATIBILITY

	Power Factor Regulator	**VARICOMP**			
	M50FA400A	M40FA630A	M40FA632A	M40FA620A	M40FA621A
MEC-20 M31FA600A	✓	✓	✓	✗	✗

8. BOARD TOPOLOGY

8.1. Terminals

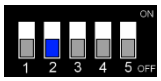
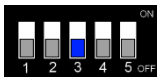
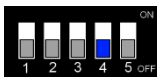
P1 P2	Power supply
U V W	Voltage sensing
A B	Current sensing
+ -	Exciter field
6 8	Analogue signal input
P Q	Connection of external potentiometer
OUT	Signal output external alarm
GND	GROUND
OB+ OB-	Connection with Varicomp board

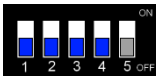


8.2. Potentiometers

VOLT		*Coarse setup of generator voltage.* Turn clockwise to increase voltage.
VFINE		*Fine setup of generator voltage (±5% of the VOLT setting).* Turn clockwise to increase voltage.
START		*Setup of the soft start ramp.* Turn clockwise to increase the ramp duration.
FREQ		*Setup of the Corner Frequency.* Turn clockwise to increase the operation frequency threshold.
SLOPE		*Setup of the Zero Volt frequency.* Turn clockwise to increase the slope of the V/f curve.
AMP		*Setup of the overexcitation threshold.* Turn clockwise to increase the limitation intervention threshold.
PAR		*Setup of the droop.* Turn clockwise to increase droop.
STAB		*Setup of regulation stability.* Turn clockwise to increase the response speed.

8.3. Dip switches

DIP SWITCHES - BLOCK A			
BLOCK A	BLOCK B	*Dip switch 1 - Selects the type of voltage sending*	
		ON:	3-phase
		OFF:	1-phases
BLOCK A	BLOCK B	*Dip switch 2 - Enables droop function (reactive droop compensation)*	
		ON:	Enabled
		OFF:	Disabled
BLOCK A	BLOCK B	*Dip switch 3 - Enables the calibration of the overexcitation limiter*	
		ON:	Limiter calibration mode enabled
		OFF:	Calibration mode disabled - Limiter entered and active
BLOCK A	BLOCK B	*Dip switch 4 - Select analogue input use mode*	
		ON:	For connection to Marelli Motori M50FA400A Power Factor Regulator
		OFF:	For connection to non Marelli Motori external devices (±5V)
Dip switch 5 not used			

DIP SWITCHES - BLOCK B						
BLOCK A	BLOCK B	*Stability parameter selection*	*1*	*2*	*3*	*4*
		Marelli Motori Generators MJB Series – Frame 160	OFF	OFF	OFF	OFF
		Marelli Motori Generators MJB Series – Frame 200	OFF	OFF	OFF	ON
		Marelli Motori Generators MJB Series – Frame 225	OFF	OFF	ON	OFF
		Marelli Motori Generators MJB Series – Frame 250	OFF	OFF	ON	ON
		Marelli Motori Generators MJB Series – Frame 315	OFF	ON	OFF	OFF
		Marelli Motori Generators MJB Series – Frame 355	OFF	ON	OFF	ON
		Marelli Motori Generators MJB Series – Frame 400	OFF	ON	ON	OFF
		Marelli Motori Generators MJB Series – Frame 450	OFF	ON	ON	ON
		Not available	ON	OFF	OFF	OFF
		Not available	ON	OFF	OFF	ON
		Not available	ON	OFF	ON	OFF
		Not available	ON	OFF	ON	ON
		Not available	ON	ON	OFF	OFF
		Not available	ON	ON	OFF	ON
		Not available	ON	ON	ON	OFF
		Not available	ON	ON	ON	ON
Dip switch 5 not used						

9. DESCRIPTION

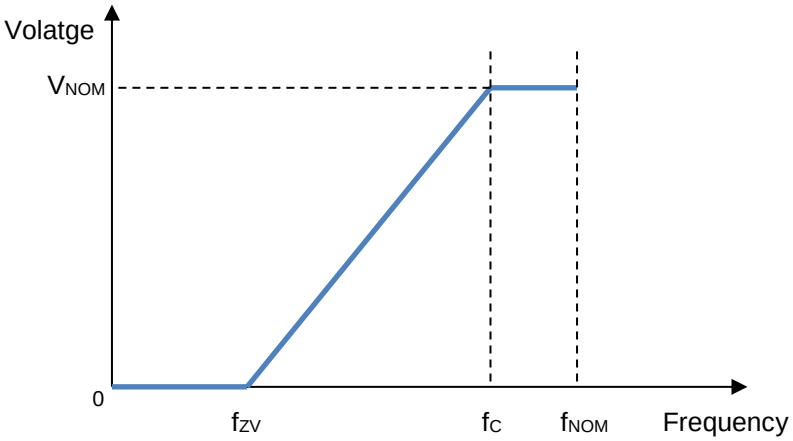
9.1. Setting generator voltage

The regulator is equipped with two single turn potentiometers for setting the generator voltage:
- ❖ VOLT, for coarse calibration of the voltage reference;
- ❖ VFINE for fine calibration of the voltage reference.

If initially positioned about midway, VFINE allows the generator voltage to be calibrated in a range of ±5% in comparison to the voltage reference set with VOLT.
The generator voltage can be further changed using the external potentiometer (see Par. 9.5).

9.2. Under frequency limitation

The regulator performs a field current reduction every time the generator is used at low speed in order to prevent overexcitation conditions that could damage the generator excitation system: in particular, the voltage reference is automatically modified and decreased according to the curve shown in the following figure.



Legenda:

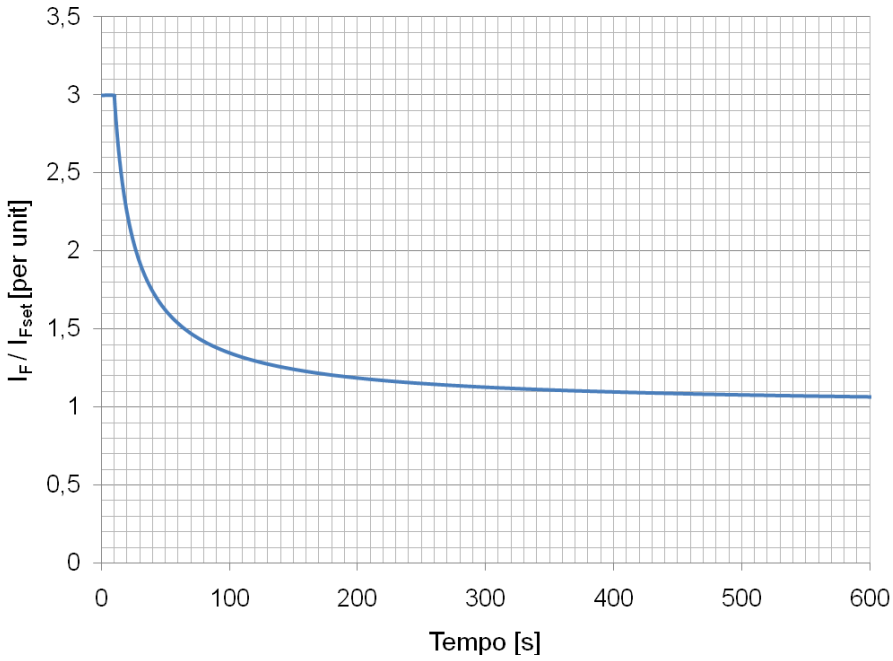
V_{NOM}	Nominal Voltage
f_{ZV}	Zero Volt Frequency
f_C	Corner Frequency
f_{NOM}	Nominal Frequency

The parameters that determine the curve and especially its slope are:
- ❖ The *Corner Frequency* (f_C), adjustable from 40 to 60 Hz using the FREQ potentiometer: it represents the frequency value below which the regulator decreases the voltage reference.
- ❖ The *Zero Volt Frequency* (f_{ZV}), adjustable from 0 to 40Hz using the SLOPE potentiometer: it represents the frequency related to the point where the reference is deleted. With f_C set, calibration of d_{ZV} determines the slope of the under frequency curve.

9.3. Droop - Reactive droop compensation

The regulator is equipped with a "Reactive Droop Compensation" function whose purpose is to allow the desired sharing of the reactive load between two or more generators operating in parallel.
When the function is enabled, through dip switch 2 - Block A (see Par. 8.3), the regulator calculates the reactive part of the generator load, starting with the voltage sensing of the generator between phases U and V and the current in phase W and consequently modifies the generate voltage reference.
A unit or capacitive power factor does not cause any change in the voltage reference. An inductive power factor causes a generator output voltage reduction (*Droop*).

9.4. Overexcitation limitation



The regulator is capable of operating in limitation of excitation current, as soon as it reaches a value which could lead to overheating the generator and, in particular, the exciter field. When this function is set properly and there is a field overcurrent, the field current value is returned to a safe value within a pre-set time interval, obtained by the curve shown in the following figure. This characteristic is calculated starting with the setting of a field current maximum value (I_{Fset}) that the regulator can continually sustain without any protection intervention.
When the field current value exceeds I_{Fset}, the field overcurrent limiter starts limiting after a time interval dependent on the field overcurrent value that was reached.
The greater the overcurrent value, the less intervention time.
The intervention consists in a decrease of the field current up to the maximum continuity value set, at which it is maintained until the following conditions occur:

- A sufficient time has passed to eliminate generator overheating.
- The operating conditions bring the excitation current value requested by the regulation to below the I_{Fset} value.

The maximum field current value that the regulator can deliver is 3 x I_{Fset} (3p.u.) for a maximum time (fixed) of 10 s. The maximum field current permitted from the regulator cannot exceed 15A, no matter the I_{Fset} setting. For example, for I_{Fset} set at 6A, the maximum deliverable current will be 15A, not 18A.

9.5. External potentiometer

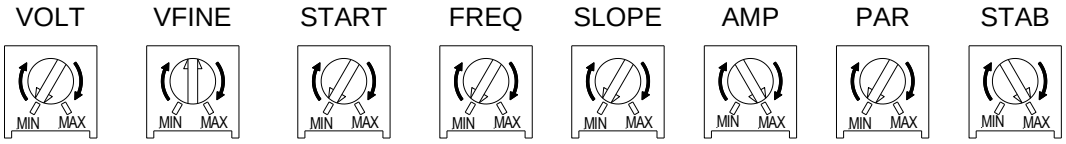
A remote generator voltage setting device (potentiometer) can be attached to terminals P and Q in place of the usually present jumper.
The external potentiometer allows the same type of calibration offered by VFINE to be performed, or ±5%,, with potentiometer 10 kΩ, 1 W minimum.
Once the external potentiometer is connected to P and Q, there will be a change in the overall voltage reference of the regulation system, for which you will need to re-set the internal VOLT trimmer of the regulator.
In particular: set the external potentiometer cursor in a middle position, after that, turn VOLT until you reach, approximately, the desired generator voltage.
At this point you can perform the fine voltage calibration using the external potentiometer.
IMPORTANT NOTE: in the event in which there is no external potentiometer or jumper attached to terminals P and Q, the regulator will not be able to deliver excitations to the generator field.

9.6. Analogue input

A control device can control the voltage reference through the terminals 6 and 8 by a dc voltage signal (Vdc). Terminal 6 represents the positive, terminal 8 represents the negative.
When dip switch 4 - BLOCK A is in the ON position, the input is suitable for use with the Power Factor Controller M50FA400A manufactured by Marelli Motori.
When dip switch 4 - BLOCK A is in OFF position, the input is suitable for use with any external device capable of providing a signal within the range +/- 5Vdc (corresponding to a voltage reference variation of +/- 15%).

10. REGULATOR SET-UP

⚠ **NOTICE: THE REGULATOR SETUP DEPENDS ON THE WIRING DIAGRAM USED. IF IT IS FORESEEN A TERMINAL BOX WIRING OTHER THAN THE ONE WITH WHICH THE ALTERNATOR IS PROVIDED, IT IS NECESSARY TO CHECK THE REGULATOR WIRING DIAGRAM, WHICH MUST COMPLY WITH THE WIRING DIAGRAMS ILLUSTRATED IN THE "DRAWINGS" CHAPTER. MOREOVER, IT IS ALWAYS NECESSARY TO REPEAT THE REGULATOR PRELIMINARY SETUP ACCORDING TO THE PROCEDURE INDICATED IN CHAPTER 10.**

Initial settings	
1	With the generator shut down, initially position the regulator potentiometers as indicated in the following figure:  where: ❖ MIN potentiometer turned completely counter clockwise: ❖ ½ potentiometer about midway; ❖ MAX potentiometer turned completely clockwise. NOTE: Whenever there is an external potentiometer connected to terminals P and Q, that potentiometer must be set midway (½).
2	The dip switches must be initially positioned, with the generator off, in relation to the type of machine and characteristics of the system. ❖ Block A: position the dip switches as indicated in the following figure:  BLOCK A in which it is assumed that the sensing is 1-phase (for 3-phase sensing, turn dip switch 1 to the ON position, see Par. 8.3). ❖ Block B: select the position of the dip switches as indicated in Par. 8.3.
3	With the generator shut down, check that there are at least one and only one of the following conditions: ❖ purple jumper between terminals P and Q; ❖ terminals P and Q connected to an external potentiometer of suitable value (see Par. 9.5).
Voltage setting	
4	Start the generator and bring it to nominal speed: the voltage at the output terminals of the generator must be less than the generator's rated voltage.
5	Turn the VOLT potentiometer clockwise until the generator output voltage reaches a value equal to about the rated voltage.
6	Using VFINE, fine calibrate the generator voltage so that it is equal to the rated voltage of the machine. Turn VFINE clockwise to increase the voltage, counter clockwise to decrease it.
Setting the underfrequency limiter	
7	Decrease the generator rotation speed to 90% of the nominal speed.
8	Turn the FREQ potentiometer clockwise until both of the following conditions occur: ❖ the red LED lights up; ❖ the generator voltage starts to decrease below the rated value of the machine.

9	Return the rotation speed to the nominal value of the machine and check that: ❖ the red LED goes out; ❖ the generator voltage is at the rated value.
NOTE: with this setting, the underfrequency limitation is calibrated so that it operates as soon as the generator speed decreases to below 90% of the nominal speed, with slope equal to about 2 per-unit (pu). For **a V/f type underfrequency limitation**, or with a slope of 1 pu, don't consider points 7 and 9, but only point 8, making sure that after the red LED lights up, FREQ is rotated counter clockwise, far enough to make the LED go out. The generator voltage must be equal to the rated voltage. Finally, turn SLOPE completely counter clockwise.	
Improvement of response speed / regulation stability	
10	❖ Improvement of response speed: turn the STAB potentiometer slightly clockwise, being sure you don't reach a generator voltage oscillation or overshoot. ❖ Improvement of regulation stability: turn the STAB potentiometer slightly counter clockwise.
11	Apply a load to the generator and check the voltage transient. Then, release the load and check the voltage transient.
12	Repeat points 10 and 11 until you reach the regulation stability or response speed desired (which is also obviously related to the type of machine and load given).
Setting the overexcitation limitation	
13	Apply a load to the generator that is equal to the machine's nominal load (nominal current and power factor).
14	Set dip switch 3 - BLOCK A to ON.
15	Turn the AMP potentiometer counter clockwise until the yellow LED lights up.
16	Turn the AMP potentiometer clockwise enough to make the yellow LED go out. The generator voltage must be equal to the rated voltage of the machine.
17	Set dip switch 3 - BLOCK A to OFF.
18	Release the load.
Setting the Reactive Droop Compensation (Droop) - Only for generator parallel	
19	Apply a load that is at least partially reactive to the generator, of known value and power factor (for example nominal current and power factor).
20	Set dip switch 2 - BLOCK A to ON.
21	Turn the PAR potentiometer clockwise until the generator voltage decreases to the desired value, depending upon the value and type of load (for example, in the event of nominal current and PF=0.8, PAR must be turned clockwise until a 4% decrease compared to nominal voltage is obtained).
22	Release the load.
Connection to Marelli Motori M50FA400A Power Factor Regulator	
23	In the event of connection to Marelli Motori M50FA400A Power Factor Regulator, set dip switch 4 - BLOCK A to ON, otherwise leave it in the OFF position.

11. TROUBLESHOOTING AND OPERATIONS

The following table completes the section *Troubleshooting and Operations* from the Marelli Motori Generator Use and Maintenance Manual, with specific attention to the aspects concerning the regulation.
It is possible that some of the problems described below can also be attributed to other generator components and not only to the regulator; it is also possible that certain regulator failures are caused by external problems or defects, for example, from the machine, the devices connected to it, incorrect use, etc.
For this reason, always consult all available documentation, in particular these User Manual, the generator Use and Maintenance Manual and the connection diagrams that are normally provided with the generator.

In the event of one of the problems described below, conduct the proper troubleshooting/problem determination by disconnecting any other devices connected to the voltage regulator (power factor regulator, overexcitation device, external remote controls, etc.).
In the event that the problem does not occur when only the voltage regulator is operating, reconnect the auxiliary devices one at a time to identify which one causes the problem. Then consult the User Manual related to the identified device.

If the available documentation is not sufficient for solving the problem, contact Marelli Motori Services for further instructions.

All operations indicated below are to be performed with the machine off, except where specifically noted.

PROBLEM	POSSIBLE CAUSE	OPERATION
With the generator started, with no-load, the voltage to the output terminals is less than the nominal value and equal to the residual voltage of the machine; this value remains even after having adjusted the VOLT and VFINE potentiometers.	There is no connection between regulator power terminals P1 and P2 and (according to the application): ❖ the auxiliary winding ❖ the main generator output terminals ❖ the PMG output	Check connections (ref. connection diagram provided with the generator).
	There is no connections between the + and - terminals on the regulator and the exciter field.	Check connections (ref. connection diagram provided with the generator).
	Terminals P and Q have no jumper or are not connected to an external potentiometer.	Check connections (ref. connection diagram provided with the generator).
	Regulator powered by auxiliary winding: the residual voltage of the auxiliary is insufficient for self-excitation (minimum voltage required, 5V).	Bring the generator to nominal speed and measure the voltage output from the auxiliary winding. If this is less than 5V, increase the residual magnetism of the generator following the instructions in the generator Use and Maintenance Manual.
	Regulator powered from main terminals: the residual generator voltage between P1 and P2 is insufficient for self-excitation (minimum voltage required, 5V).	Bring the generator to nominal speed and measure the voltage between P1 and P2. If this is less than 5V, increase the residual magnetism of the generator following the instructions in the generator Use and Maintenance Manual.
	Regulator powered from the PMG: the PMG output voltage and P1 and P2 input voltage is insufficient for self-excitation (minimum voltage required, 5V).	Bring the generator to nominal speed and measure the voltage PMG output voltage. If it is less than 100V, contact Marelli Motori Services for further instructions.
	Internal fuse interrupted	Replace the fuse.
	Regulator failure.	Replace the regulator.
Once the blown fuse is replaced, there is again a fuse interruption.	Regulator failure.	Replace the regulator.

PROBLEM	POSSIBLE CAUSE	OPERATION
With the generator on, with no load, the voltage at the output terminals is greater than the nominal value and cannot be changed using the VOLT and VFINE potentiometers.	With dip switch 1 - Block A set to OFF, one or both of regulator terminals U and V are not connected to the respective generator phases.	Check connections (ref. connection diagram provided with the generator).
	With dip switch 1 - Block A set to ON, one or more of regulator terminals U, V and W are not connected to the respective generator phases.	Check connections (ref. connection diagram provided with the generator).
	With 1-phase sensing (only terminals U and V used), dip switch 1 - Block A is in the ON position.	Set dip switch 1 - BLOCK A to OFF.
	Regulator failure.	Replace the regulator.
With the generator started and operating off-grid, the generator voltage decreases to below the nominal value as soon as a load is applied or if the load is increased (within the generator's nom. limits).	LED lit: overexcitation limitation is active even if the load applied is lower than the nominal load.	Perform the correct overexcitation limitation set-up, according to the instructions in the *Regulator Set-up* paragraph in this User Manual.
	Droop function active: dip switch 2 - Block A is in the ON position.	Set dip switch 2 - BLOCK A to OFF.
	Regulator failure.	Replace the regulator.
Generator parallel operation - incorrect distribution of reactive power: with the generator started and off the grid, loaded with a partially reactive load, the voltage does not decrease to below the nominal value of the machine.	Droop function not active: dip switch 2 - Block A is in the OFF position.	Set dip switch 2 - BLOCK A to ON.
	Droop function not set: dip switch 2 - Block A is in the ON position but the PAR potentiometer is in the MIN posit.	Perform the correct droop set-up (PAR potent.), according to the *Regulator Set-up* paragraph in this User Manual.
	The measurement current transformer (CT) linked to terminals A and B is backwards.	Reverse the CT conductors on terminals A and B.
	The measurement current transformer (CT) linked to terminals A and B is short-circuited or not connected.	Check connections (ref. connection diagram provided with the generator).
	Voltage and current sensing incorrect. Sensing is correct if: ❖ Voltage: phases U and V are connected to regulator terminals U and V; ❖ Current: measurem. current transformer is on phase W.	Check connections (ref. connection diagram provided with the generator).
	Regulator failure.	Replace the regulator.
Parallel network operation (in connection with M50FA400A board): incorrect, insufficient or absent power factor (or reactive power) regulation.	Incorrect selection of the analogue input: dip switch 4 - Block A is OFF.	Set dip switch 4 - Block A to ON.
	No connection between regulator terminals 6 and 8 and the M50FA400A power factor regulator.	Check connections (ref. connection diagram provided with the generator).
	Connections to terminal 6 and 8 are reversed.	Check connections (ref. connection diagram provided with the generator).
	Regulator failure.	Replace the regulator.

12. AVR ANALYSIS

The following paragraph provides a procedure for the identification of the AVR failure and the proper filling in of the annexed report form ((APPENDIX)).

All the checks and measurements specified herein must ALWAYS be performed with the AVR disconnected from the alternator.

This document allows to identify the failures occurring in the AVR envelope, in the main components and/or in the AVR power supply circuit.
The accuracy and steadiness of the regulation can only be verified in the final application for which the alternator is used.

IMPORTANT: in case of malfunction or if any AVR failure is detected, for any AVR claim to be accepted the user must have carried out the preliminary procedure described herein and filled in the report form in its entirety.

In case you experience any of the difficulties listed in this chapter please follow the troubleshooting and repair procedure described

The report form includes the following sections:

#1 - GENERAL INFORMATION

Specify the serial numbers of the alternator and of the AVR. The serial number and the part number of the AVR are specified in a label affixed on the AVR plastic support. The User must also provide additional information, like the failure date, the site and the AVR operating time.

#2 - VISUAL / MULTI-METER CHECKS

The User must perform a visual inspection of the AVR and all the electrical checks described in the following pages. If the result of any check is positive, tick the relevant box. If the result is negative, leave the box blank.

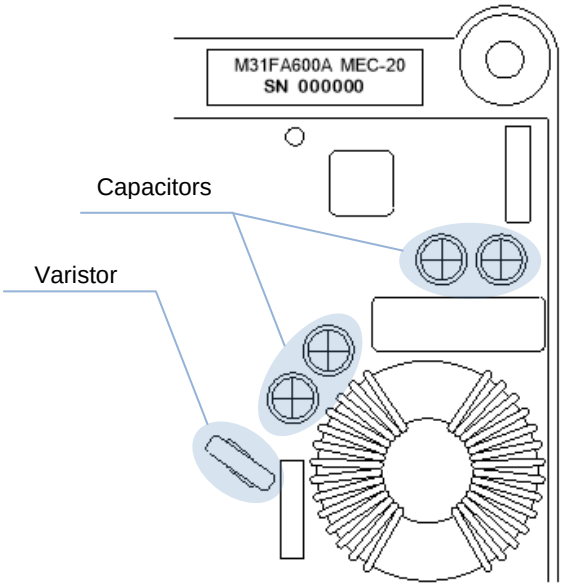
#3 - DETECTED PROBLEMS

Select one or more options from the possible failure mode list. Also add any further information about the failure, if available. In this section the User must describe what is detected during the user alternator test.

#4 - ADDITIONAL INFORMATION

Enter here a description of the application and/or any additional information helping to define the operating conditions under which the failure occurred.

#2 - VISUAL / MULTI-METER CHECKS
CHECK A – Varistor/Capacitors
(Varistor/Capacitors)



Visual inspection:
- Check that the varistor and the capacitors are not damaged.
  Damage types may be:
  1- Exploded/blown component.
  2- Broken component.

Multi-meter check:
- None.

Check result:
- If the result of all the above checks has been positive, tick the **ok** box in section #2 of the report form, Check A.
  Otherwise leave the box blank.

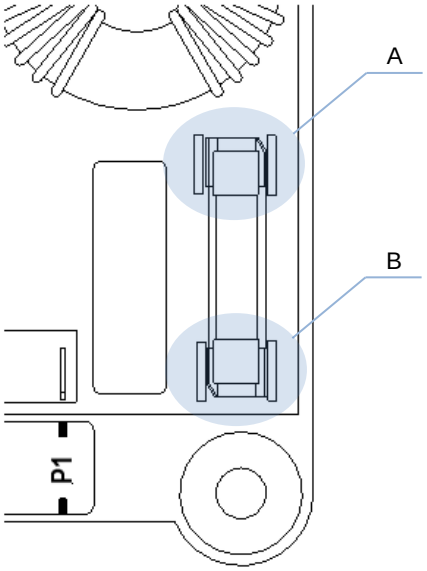
In case of component failure:

Possible effects on regulation:
- The alternator output voltage is lower than the rated voltage and equal to the residual one.
  You cannot set the proper value using the VOLT trimmer.
- None.

Possible failure causes:
- Mechanical damage due to impact/shock.
- Voltage exceeding 420Vrms between terminals P1 and P2 due to alternator overvoltage.
- Improper connection, which resulted in a voltage higher than 420Vrms between terminals P1 and P2.

Actions:
- Replace the AVR.

#2 - VISUAL / MULTI-METER CHECKS
CHECK B – Fuse
(Fuse)



Visual inspection:
- Check that the fuse is not broken or blown.
- Check that the fuse holder blades are tightly in place around fuse terminals A and B.
- Check whether there is any resin and/or dirt between the fuse holder and the fuse terminals.

Multi-meter check:
1- Remove the fuse from the fuse holder.
2- Select the resistance measurement mode on the multi-meter.
3- Connect the multi-meter probes to pins A and B of the fuse and check that the measured resistance is < 1Ω. If it is not, the fuse has to be replaced.

Check result:
- If the result of all the above checks has been positive, tick the **ok** box in section #2 of the report form, Check B.
  Otherwise leave the box blank.

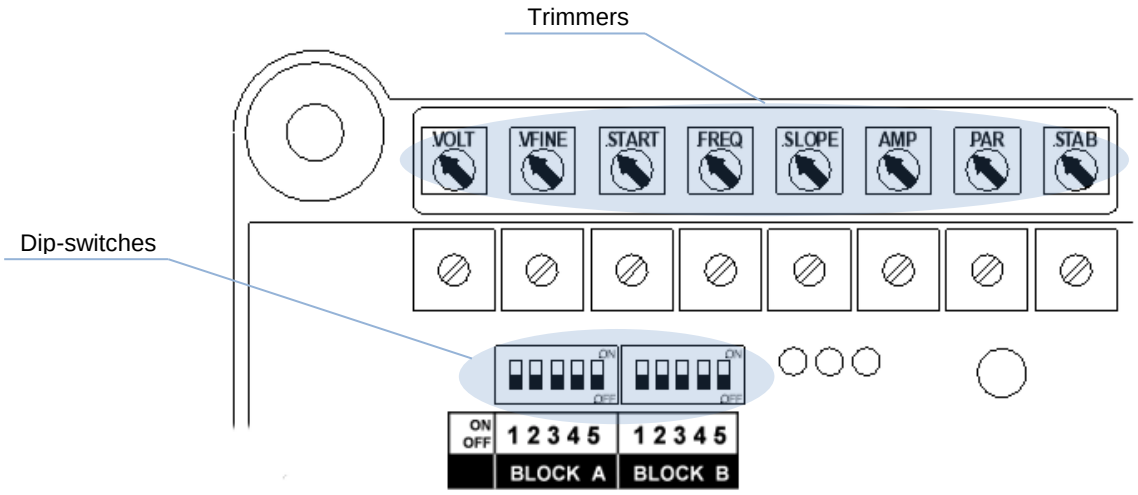
In case of component failure:

Possible effects on regulation:
- The alternator output voltage is lower than the rated voltage and equal to the residual one. You cannot set the proper value using the VOLT trimmer.

Possible failure causes:
- Mechanical damage due to impact/shock.
- Presence of resin on the fuse terminals and/or on the fuse holder blades.
- Not enough contact between the fuse and the fuse-holder.
- Improper connection.
- Exciter failure: check the insulation of the exciter with a megaohmmeter (insulation is appropriate if resistance is >30MΩ).
- Short circuit between terminals + and -.

Actions:
- Replace the fuse and try restarting the alternator. In case of any further fuse failure, replace the AVR.

#2 - VISUAL / MULTI-METER CHECKS
CHECK C –Trimmers/Dips
(Trimmer/Dip switches)



Visual inspection:
- The AVR trimmers and dip-switches must show no sign of mechanical damage; in addition, they must not be covered with resin.

Multi-meter check:
- None.

Check result:
- If the result of all the above checks has been positive, tick the **ok** box in section #2 of the report form, Check C. Otherwise leave the box blank.

In case of component failure:

Possible effects on regulation:
- If the trimmer is covered with resin or damaged, the trimmer rotor can no longer be operated for any setting.
- If the dip-switch is covered with resin or damaged, it might be impossible to have the proper stability set-up for voltage adjustment.

Possible failure causes:
- Mechanical damage due to impact/shock.
- Improper AVR resin application process.

Actions:
- Replace the AVR.

#2 - VISUAL / MULTI-METER CHECKS
CHECK D –Box/Terminals
(Plastic support/Terminals)

Visual inspection:
- The AVR must not show any sign of mechanical damage, or any broken component; in addition, terminals must not be covered with resin.

Multi-meter check:
- None.

Check result:
- If the result of all the above checks has been positive, tick the **ok** box in section #2 of the report form, Check D.
- Otherwise leave the box blank.

In case of component failure:

Possible effects on regulation:
- If the plastic support is broken, the AVR cannot be reliably and safely installed inside the terminal box.
- If terminals are covered with resin or damaged, it might be impossible to establish a proper connection with the alternator. This may result in various AVR operating faults, depending on the terminals concerned.

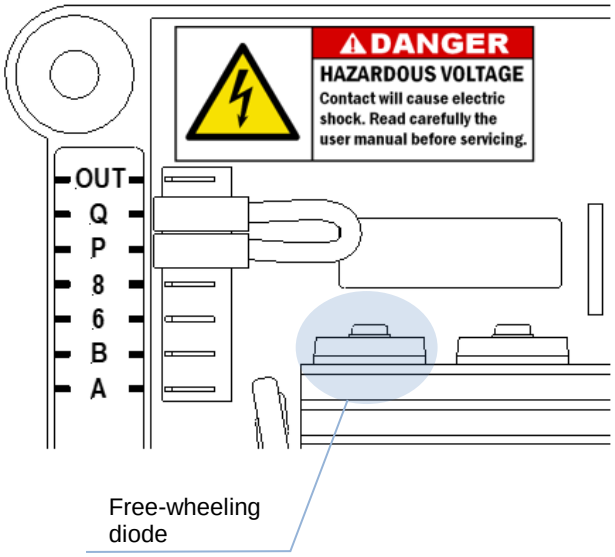
Possible failure causes:
- Mechanical damage due to impact/shock.
- Improper AVR resin application process.

Actions:
- Replace the AVR.

#2 - VISUAL / MULTI-METER CHECKS
CHECK E – Free-wheeling diode
(Free wheeling diode)

Visual inspection:
- Check that the diode shown in the figure is not damaged.
  Possible damage types are:
  1- Diode blown.
  2- Diode broken.

Multi-meter check:
1- In the multi-meter, select the diode measurement mode.
2- Connect the multi-meter probes to terminals - and + of the AVR.
   Expected values are:
   COM probe on +: 0.5 to 0.7V.
   COM probe on -: open (OL).
   If at least one of the above measurement is not met, the free- wheeling diode is damaged.



Check result:
- If the result of all the above checks has been positive, tick the **ok** box in section #2 of the report form, Check E.
  Otherwise leave the box blank.

In case of component failure:

Possible effects on regulation:
- The alternator output voltage is lower than the rated voltage and equal to the residual one.
- You cannot set the proper value using the VOLT trimmer.
- The alternator output voltage is higher than the rated voltage.
- You cannot set the proper value using the VOLT trimmer.

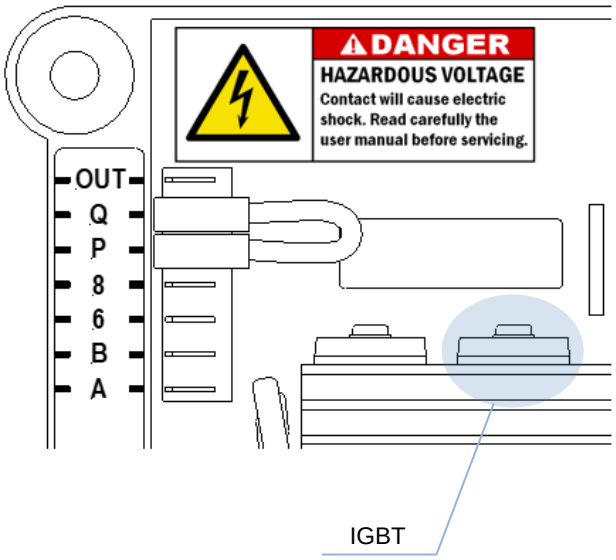
Possible failure causes:
- Mechanical damage due to impact/shock.
- The alternator output voltage is higher than the rated voltage.
- You cannot set the proper value using the VOLT trimmer.
- Not enough contact between the AVR terminals and the FASTON connectors, resulting in sparks and voltage peaks.

Actions:
- Replace the AVR.

#2 - VISUAL / MULTI-METER CHECKS
CHECK F – IGBT

Visual inspection:
- Check that the IGBT shown in the figure is not damaged.
  Possible damage types are:
  1- IGBT blown.
  2- IGBT broken.

Multi-meter check:
1- In the multi-meter, select the resistance measurement mode.
2- Connect the multi-meter probes to terminals - and Q of the AVR.
  Expected values are:
  COM probe on Q: 100 to 150kΩ.
  COM probe on -:100 to 150kΩ.
  If any of the above measurements is not met, the IGBT is damaged.



Check result:
- If the result of all the above checks has been positive, tick the **ok** box in section #2 of the report form, Check F.
  Otherwise leave the box blank.

In case of component failure:

Possible effects on regulation:
- The alternator output voltage is lower than the rated voltage and equal to the residual one.
  You cannot set the proper value using the VOLT trimmer.
- The alternator output voltage is higher than the rated voltage.
  You cannot set the proper value using the VOLT trimmer.

Possible failure causes:
- Mechanical damage due to impact/shock.
- Improper connection on terminals + and -.
- Short circuit between terminals + and -.
- Not enough contact between the AVR terminals and the FASTON connectors, resulting in sparks and voltage peaks.

Actions:
- Replace the AVR.

13. MAINTENANCE

The only preventative maintenance required for the regulator is to check the connections between the regulator itself and the system: make sure they are clean and tight and that the cabling shows no imperfections or damage.
The M31FA600A is a surface mounting (SMD) circuit board protected by a polyurethane resin that protects the device from dampness, dust and aggressive environments: in case of malfunction or damage of any type, any modifications, repairs or adaptations not previously approved by Marelli Motori S.p.A are forbidden.

14. ASSISTANCE

For any questions regarding the connection diagrams, information or any board malfunction, damage or problem, contact Marelli Motori Services.